KIET Deemed to be University, Ghaziabad

Assignment: UML Component Diagram

Name: Ashutosh Tiwari | Branch: CSE | Section: B | Semester: V

University Reg. No.: 202401100200098 | University Roll No.: 2400290100098 | Project: Digital Payment Gateway Simulation

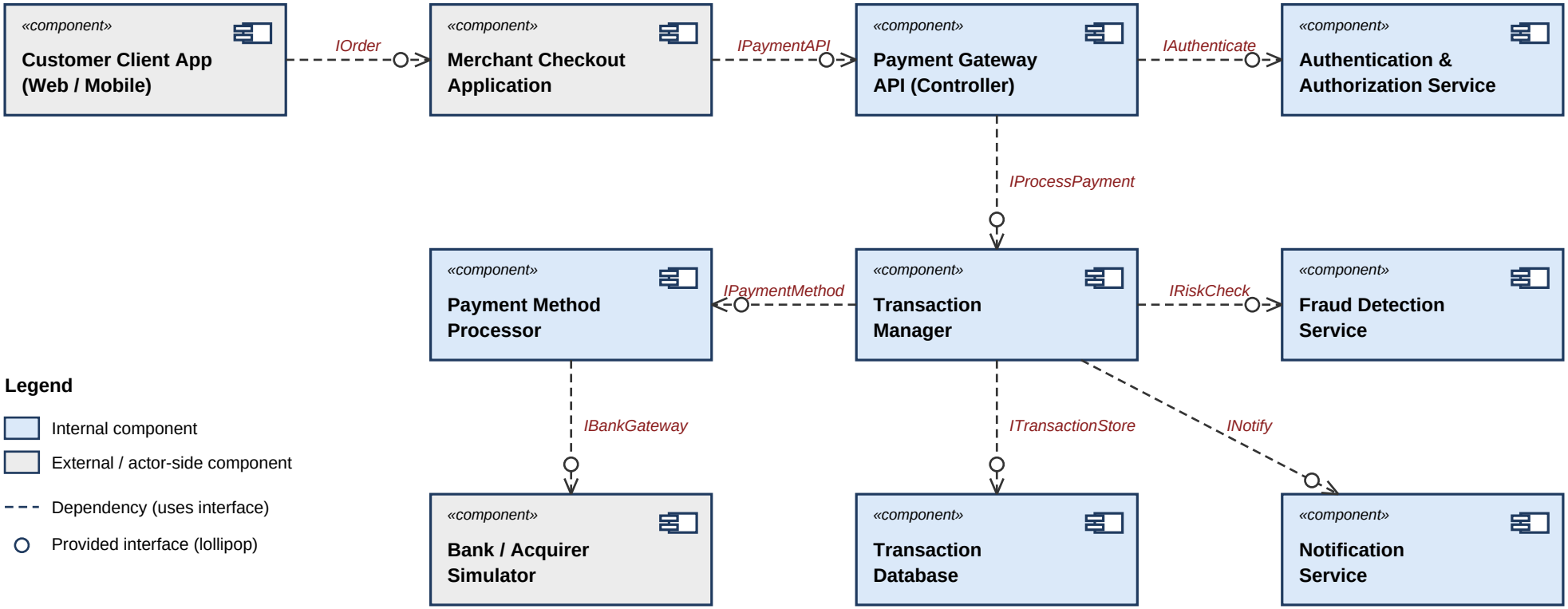

Fig.: Component Diagram of Digital Payment Gateway Simulation

KIET Deemed to be University, Ghaziabad

Assignment: UML Component Diagram - Component Description

Name: Ashutosh Tiwari Branch: CSE Section: B Semester: V
University Reg. No.: 202401100200098 University Roll No.: 2400290100098 Project: Digital Payment Gateway Simulation

Component	Responsibility	Provided Interface	Required Interface
Customer Client App	UI where the customer selects items, enters payment details and views the result	-	IOrder
Merchant Checkout Application	Creates the order and forwards the payment request to the gateway	IOrder	IPaymentAPI
Payment Gateway API	Single entry point; validates requests and routes them to internal services	IPaymentAPI	IAuthenticate, IProcessPayment
Authentication & Authorization Service	Verifies merchant API keys, user credentials and OTP/PIN	IAuthenticate	-
Transaction Manager	Controls the payment life cycle (initiated, authorised, success, failed, refunded)	IProcessPayment	IRiskCheck, IPaymentMethod, ITransactionStore, INotify
Fraud Detection Service	Applies risk rules (amount limit, repeated attempts) before approval	IRiskCheck	-
Payment Method Processor	Handles Card, UPI, Net Banking and Wallet payment logic	IPaymentMethod	IBankGateway
Bank / Acquirer Simulator	Simulates the bank: balance check, debit/credit and approve/decline response	IBankGateway	-
Transaction Database	Stores transactions, payment status and receipts	ITransactionStore	-
Notification Service	Sends SMS/e-mail confirmation of success or failure to customer and merchant	INotify	-